



INSTALLATION GUIDE

*A guide for installing or migrating to CircleCI Server
v3.4.8 on Amazon Web Services*

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Version 3.4.8, 02/02/2024: FINAL

CircleCI Server v3.x Installation Phase 1	1
Phase 1: Prerequisites	1
Install required software	1
AWS required software	2
S3 compatible storage required software	2
Create a Kubernetes cluster	2
EKS	3
Create a new GitHub OAuth app	4
Frontend TLS certificates	6
AWS Route 53	6
Encryption/signing keys	7
Artifact signing key	7
Encryption signing key	7
Object storage and permissions	7
Create an S3 storage bucket	8
CircleCI Server v3.x Installation Phase 2	14
Phase 2: Core services installation	14
Installing behind an HTTP Proxy (optional)	15
Frontend Settings	15
Using ACM TLS certificates	17
Encryption	17
GitHub	18
Object storage	19
S3 compatible	19
Postgres, MongoDB, Vault settings	20
Save and deploy	20
Create DNS entry	20
Validation	21
CircleCI Server v3.x Installation Phase 3	22
Phase 3: Execution environment installation	22
Output Processor	22
Nomad Clients	23
Cluster Creation with Terraform	23
Nomad Autoscaler	25
Configure and Deploy	26
Nomad Clients Validation	27
VM service	28
AWS	28
VM Service Validation	32
Runner	32
Overview	32

CircleCI Server v3.x Installation Phase 4	33
Phase 4: Post installation	33
Set up backup and restore	33
Server 3.x backups on AWS	33
Step 1 - Create an AWS S3 bucket	33
Step 2 - Set up permissions for Velero	34
Step 3 - Install and start Velero	35
Server 3.x backups with S3 Compatible Storage	36
Step 1 - Configure mc client	36
Step 2 - Create a bucket	36
Step 3 - Create a user and policy	37
Step 4 - Install and start Velero	38
Creating backups	38
Option 1 - Create a backup with KOTS CLI	39
Option 2 - Create a backup with KOTS Admin Console	39
Orbs	39
Email Notifications	40
CircleCI Server v3.x Migration	41
Prerequisites	41
External Datastores Only	41
Internal Datastore Only	41
Migration	41
Step 1 - Clone the repository and run the migration script	42
Step 2 - Validate your migration to Server 3.0	43
Step 3 - Update your team	43
Frequently Asked Questions	43
Where did all my job and build history go?	43
Why does nothing happen when I select "Start Building" on my project after migration?	43
I got an error "Error from server (NotFound):"	43
Transitioning to pipelines	44
CircleCI Server v3.x Hardening Your Cluster	45
Network Topology	45
Network Traffic	45
Kubernetes Load Balancers	46
Ingress	46
Egress	46
Common Rules for Compute Instances	46
Ingress	46
Egress	46
Kubernetes Nodes	47
Intra-node traffic	47

Ingress	47
Egress	47
Nomad Clients	47
Ingress	47
Egress	48
External VMs	48
Ingress	48
Egress	48

CircleCI Server v3.x Installation Phase 1

Phase 1: Prerequisites

CircleCI server v3.x is installed in 4 phases. There is a validation step at the end of each phase, allowing you to confirm success before moving to the next phase. Depending on your requirements, phases 3 and 4 may include multiple steps. This installation guide assumes you have already read the [CircleCI Server v3.x Overview](#).

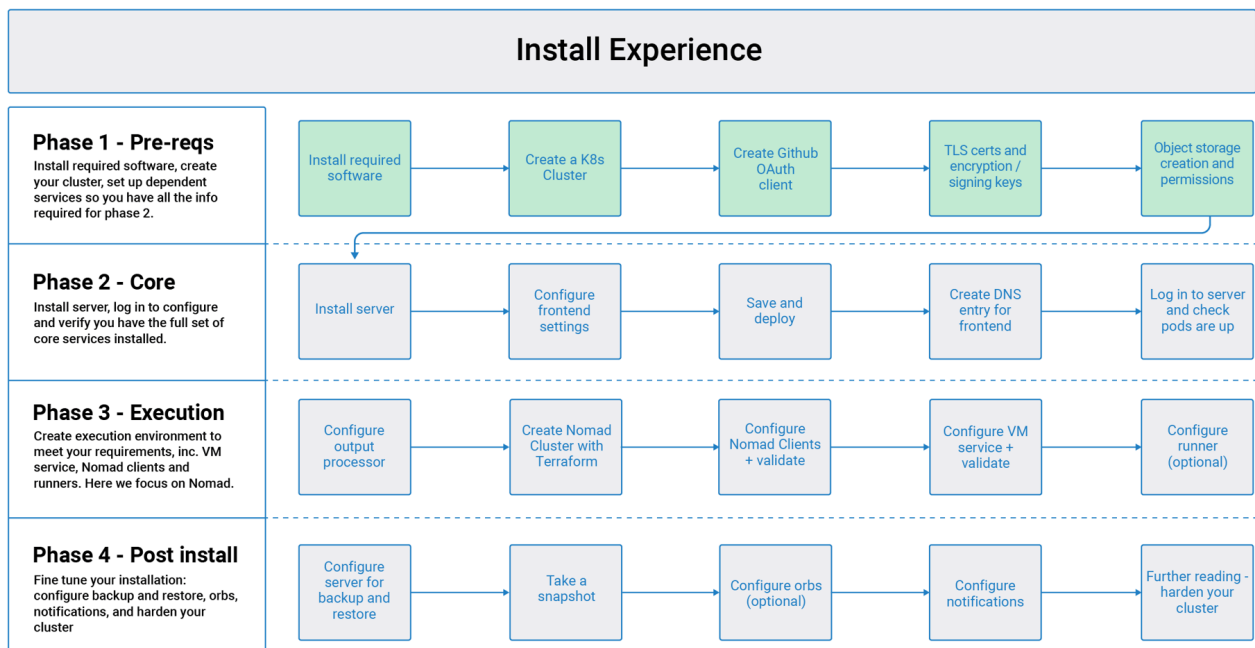


Figure 1. Installation Experience Flow Chart Phase 1



In the following sections, replace any items or credentials displayed between `< >` with your details.

Install required software

Download and install the following software before continuing:

Tool	Version	Used for	Notes
Terraform	0.15.4 or greater	Infrastructure Management	
kubectl	1.19 or greater	Kubernetes CLI	
Helm	3.9.2 or greater	Kubernetes Package Management	

Tool	Version	Used for	Notes
KOTS: Mac or Linux .	1.65.0 *	Replicated Kubernetes Application Management. KOTS is a kubectl plugin .	Once you have extracted kots from the tar.gz (<code>tar zxvf kots_linux_amd64.tar.gz</code>), run <code>sudo mv kots /usr/local/bin/kubectl-kots</code> to install it. Mac users will need to grant a security exception.
Velero CLI	Latest	Backup and restore capability	See Velero's supported providers documentation for further information.

* Please take note of the supported KOTS versions for your Kubernetes cluster. [KOTS version compatibility](#)

AWS required software

- [AWS CLI](#)

S3 compatible storage required software

- Install and configure [MinIO CLI](#) for your storage provider.

Create a Kubernetes cluster

CircleCI server installs into an existing Kubernetes cluster. The application uses a large number of resources. Depending on your usage, your Kubernetes cluster should meet the following requirements:

Number of daily active CircleCI users	Minimum Nodes	Total CPU	Total RAM	NIC speed
< 500	3	12 cores	32 GB	1 Gbps
500+	3	48 cores	240 GB	10 Gbps

Supported Kubernetes versions:

CircleCI Version	Kubernetes Version
3.0.0 - 3.2.1	< 1.21
3.2.2 - 3.3.0	1.16 - 1.21
3.4.0 - 3.4.6	1.16 - 1.23

Creating a Kubernetes cluster is your responsibility. Please note:

- Your cluster must have outbound access to pull Docker containers and verify your license. If you do not want to provide open outbound access, see our [list of ports](#) that need access.
- You must have appropriate permissions to list, create, edit, and delete pods in your cluster. Run this command to verify your permissions:

```
kubectl auth can-i <list|create|edit|delete> pods
```

- There are no requirements regarding VPC setup or disk size for your cluster. It is recommended that you set up a new VPC rather than use an existing one.

EKS

You can learn more about creating an Amazon EKS cluster [here](#). We recommend using `eksctl` to create your cluster, which creates a VPC and selects the proper security groups for you.

1. [Install](#) and [configure](#) the AWS CLI for your AWS account.
2. Install `eksctl`.
3. Create your cluster by running the following (Cloud formation with `eksctl` and EKS can take more than 20 minutes to complete):

```
eksctl create cluster --name=circleci-server --nodes 4 --node-type m5.xlarge
```

4. Once the cluster has been created, you can use the following command to configure `kubectl` access:

```
eksctl utils write-kubeconfig --cluster circleci-server
```



You may see the following error `AWS STS Access - cannot get role ARN for current session: InvalidClientTokenID`. This means your AWS credentials are invalid, or your IAM user does not have permission to create an EKS cluster. Proper IAM permissions are necessary to use `eksctl`. See the AWS documentation on [IAM permissions](#).

Enable Workload Identities in GKE (optional)

[Workload Identities](#) for GKE allow workloads/pods in your GKE cluster to impersonate IAM service accounts to access Google Cloud services without using static service account credentials. In order to use Workload Identities you must enable them on your GKE cluster.

1. Enable Workload Identity on existing cluster

```
gcloud container clusters update "<CLUSTER_NAME>" \
  --region="<REGION>" \
  --workload-pool="<PROJECT_ID>.svc.id.goog"
```

2. Get node pools of existing GKE cluster

```
gcloud container node-pools list --cluster "<CLUSTER_NAME>" --region "<REGION>"
```

3. Update existing node pools

```
gcloud container node-pools update "<NODEPOOL_NAME>" \
  --cluster="<CLUSTER_NAME>" \
  --workload-metadata="GKE_METADATA" \
  --region="<REGION>"
```

You must repeat Step 3 for all the existing node pools. Follow these links for steps to enable Workload Identity for your Kubernetes service accounts: [Nomad Autoscaler](#), [VM](#) and [Object-Storage](#)

Create a new GitHub OAuth app



If GitHub Enterprise and CircleCI server are not on the same domain, then images and icons from GHE will fail to load in the CircleCI web app.

Registering and setting up a new GitHub OAuth app for CircleCI server allows for authorization control to your server installation using GitHub OAuth and for updates to GitHub projects/repos using build status information.

1. In your browser, navigate to **your GitHub instance > Settings > Developer Settings > OAuth Apps** and click the **New OAuth App** button.

Register a new OAuth application

Application name *

circleci-server

Something users will recognize and trust.

Homepage URL *

https://circleci.someplace.com

The full URL to your application homepage.

Application description

Application description is optional

This is displayed to all users of your application.

Authorization callback URL *

https://circleci.someplace.com/auth/github

Your application's callback URL. Read our [OAuth documentation](#) for more information.

Register application

Cancel

Figure 2. New GitHub OAuth App

2. Complete the following fields, based on your planned installation:
 - **Homepage URL:** The URL of your planned CircleCI installation.
 - **Authorization callback URL:** The authorization callback URL is the URL of your planned CircleCI installation followed by `/auth/github`
3. Once completed, you will be shown the **Client ID**. Select **Generate a new Client Secret** to generate a Client Secret for your new OAuth App. You need these values when you configure CircleCI server.

circleci-server

 nathanfishcircleci owns this application.

Transfer ownership

You can list your application in the [GitHub Marketplace](#) so that other users can discover it.

List this application in the Marketplace

0 users

Revoke all user tokens

Client ID

1d2249984d6ce6a5d599

Client secrets

Generate a new client secret

You need a client secret to authenticate as the application to the API.

Figure 3. Client ID and Secret



If using GitHub Enterprise, you also need a personal access token and the domain name of your GitHub Enterprise instance.

Frontend TLS certificates

By default, CircleCI server creates self-signed certificates to get you started. In production, you should supply a certificate from a trusted certificate authority. The [Let's Encrypt](#) certificate authority, for example, can issue a free certificate using their [certbot](#) tool. The sections below cover using Google Cloud DNS and AWS Route 53.

AWS Route 53

1. If you are using AWS Route 53 for DNS, you need the **certbot-route53** plugin installed. You can install the plugin with the following command:

```
pip3 install certbot-dns-route53
```

2. Then execute this example to create a private key and certificate (including intermediate certificates) locally in `/etc/letsencrypt/live/<CIRCLECI_SERVER_DOMAIN>`:

```
certbot certonly --dns-route53 -d "<CIRCLECI_SERVER_DOMAIN>" -d "app.<CIRCLECI_SERVER_DOMAIN>"
```



It is important that your certificate contains both your domain and the `app.*` subdomain as subjects. For example, if you host your installation at `server.example.com`, your certificate must cover `app.server.example.com` and

server.example.com.

You will need these certificates later, and they can be retrieved locally with the following commands:

```
ls -l /etc/letsencrypt/live/<CIRCLECI_SERVER_DOMAIN>
```

```
cat /etc/letsencrypt/live/<CIRCLECI_SERVER_DOMAIN>/fullchain.pem
```

```
cat /etc/letsencrypt/live/<CIRCLECI_SERVER_DOMAIN>/privkey.pem
```

Encryption/signing keys

These keysets are used to encrypt and sign artifacts generated by CircleCI. You need these values to configure server.



Store these values securely. If they are lost, job history and artifacts will not be recoverable.

Artifact signing key

To generate an artifact signing key, run the following command:

```
docker run circleci/server-keysets:latest generate signing -a stdout
```

Encryption signing key

To generate an encryption signing key, run the following command:

```
docker run circleci/server-keysets:latest generate encryption -a stdout
```

Object storage and permissions

Server 3.x hosts build artifacts, test results, and other state object storage. We support the following:

- [AWS S3](#)
- [MinIO](#)
- [Google Cloud Storage](#)

While any S3 compatible object storage may work, we test and support AWS S3 and MinIO.

Please choose the option that best suits your needs. A Storage Bucket Name is required, in addition to the fields listed below, depending on whether you are using AWS or GCP. Before proceeding,

ensure the bucket name you provide exists in your chosen object storage provider.



If you are installing behind a proxy, object storage should be behind this proxy also. Otherwise proxy details will need to be supplied at the job level within every project `.circleci/config.yml` to allow artifacts, test results, cache save and restore, and workspaces to work. For more information see the [Configuring a Proxy](#) guide.

Create an S3 storage bucket

You will need the following details when you configure CircleCI server.

- **Storage Bucket Name** - The bucket name to be used for server.
- **Access Key ID** - Access Key ID for S3 bucket access.
- **Secret Key** - Secret Key for S3 bucket access.
- **AWS S3 Region** - AWS region of bucket, if your provider is AWS. You will either have an AWS region or S3 Endpoint, depending on your specific setup.
- **S3 Endpoint** - API endpoint of S3 storage provider, when your storage provider is not Amazon S3.

Steps to create your S3 bucket:

1. Create AWS S3 Bucket

```
aws s3api create-bucket \
  --bucket <YOUR_BUCKET_NAME> \
  --region <YOUR_REGION> \
  --create-bucket-configuration LocationConstraint=<YOUR_REGION>
```



`us-east-1` does not support a LocationConstraint. If your region is `us-east-1`, omit the bucket configuration

2. Create an IAM user for CircleCI server

```
aws iam create-user --user-name circleci-server
```

3. Create a policy document *policy.json*

If using IAM Roles for Service Accounts (IRSA) for authentication, use the following content

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
```

```
"s3:PutAnalyticsConfiguration",
"s3:GetObjectVersionTagging",
"s3:CreateBucket",
"s3:GetObjectAcl",
"s3:GetBucketObjectLockConfiguration",
"s3:DeleteBucketWebsite",
"s3:PutLifecycleConfiguration",
"s3:GetObjectVersionAcl",
"s3:PutObjectTagging",
"s3:DeleteObject",
"s3:DeleteObjectTagging",
"s3:GetBucketPolicyStatus",
"s3:GetObjectRetention",
"s3:GetBucketWebsite",
"s3:GetJobTagging",
"s3:DeleteObjectVersionTagging",
"s3:PutObjectLegalHold",
"s3:GetObjectLegalHold",
"s3:GetBucketNotification",
"s3:PutBucketCORS",
"s3:GetReplicationConfiguration",
"s3:ListMultipartUploadParts",
"s3:PutObject",
"s3:GetObject",
"s3:PutBucketNotification",
"s3:DescribeJob",
"s3:PutBucketLogging",
"s3:GetAnalyticsConfiguration",
"s3:PutBucketObjectLockConfiguration",
"s3:GetObjectVersionForReplication",
"s3:GetLifecycleConfiguration",
"s3:GetInventoryConfiguration",
"s3:GetBucketTagging",
"s3:PutAccelerateConfiguration",
"s3:DeleteObjectVersion",
"s3:GetBucketLogging",
"s3:ListBucketVersions",
"s3:ReplicateTags",
"s3:RestoreObject",
"s3:ListBucket",
"s3:GetAccelerateConfiguration",
"s3:GetBucketPolicy",
"s3:PutEncryptionConfiguration",
"s3:GetEncryptionConfiguration",
"s3:GetObjectVersionTorrent",
"s3:AbortMultipartUpload",
"s3:PutBucketTagging",
"s3:GetBucketRequestPayment",
"s3:GetAccessPointPolicyStatus",
"s3:GetObjectTagging",
"s3:GetMetricsConfiguration",
```

```

        "s3:PutBucketVersioning",
        "s3:GetBucketPublicAccessBlock",
        "s3:ListBucketMultipartUploads",
        "s3:PutMetricsConfiguration",
        "s3:PutObjectVersionTagging",
        "s3:GetBucketVersioning",
        "s3:GetBucketAcl",
        "s3:PutInventoryConfiguration",
        "s3:GetObjectTorrent",
        "s3:PutBucketWebsite",
        "s3:PutBucketRequestPayment",
        "s3:PutObjectRetention",
        "s3:GetBucketCORS",
        "s3:GetBucketLocation",
        "s3:GetAccessPointPolicy",
        "s3:GetObjectVersion",
        "s3:GetAccessPoint",
        "s3:GetAccountPublicAccessBlock",
        "s3:ListAllMyBuckets",
        "s3:ListAccessPoints",
        "s3:ListJobs"
    ],
    "Resource": [
        "arn:aws:s3:::<YOUR_BUCKET_NAME>",
        "arn:aws:s3:::<YOUR_BUCKET_NAME>/*"
    ]
},
{
    "Effect": "Allow",
    "Action": [
        "iam:GetRole",
        "sts:AssumeRole"
    ],
    "Resource": "<YOUR_OBJECT_STORAGE_ROLE>"
}
]
}

```

Otherwise, if using IAM keys for authentication, use the following content

```

{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Effect": "Allow",
            "Action": [
                "s3:PutAnalyticsConfiguration",
                "s3:GetObjectVersionTagging",
                "s3:CreateBucket",
                "s3:GetObjectAcl",

```

"s3:GetBucketObjectLockConfiguration",
"s3:DeleteBucketWebsite",
"s3:PutLifecycleConfiguration",
"s3:GetObjectVersionAcl",
"s3:PutObjectTagging",
"s3:DeleteObject",
"s3:DeleteObjectTagging",
"s3:GetBucketPolicyStatus",
"s3:GetObjectRetention",
"s3:GetBucketWebsite",
"s3:GetJobTagging",
"s3:DeleteObjectVersionTagging",
"s3:PutObjectLegalHold",
"s3:GetObjectLegalHold",
"s3:GetBucketNotification",
"s3:PutBucketCORS",
"s3:GetReplicationConfiguration",
"s3:ListMultipartUploadParts",
"s3:PutObject",
"s3:GetObject",
"s3:PutBucketNotification",
"s3:DescribeJob",
"s3:PutBucketLogging",
"s3:GetAnalyticsConfiguration",
"s3:PutBucketObjectLockConfiguration",
"s3:GetObjectVersionForReplication",
"s3:GetLifecycleConfiguration",
"s3:GetInventoryConfiguration",
"s3:GetBucketTagging",
"s3:PutAccelerateConfiguration",
"s3:DeleteObjectVersion",
"s3:GetBucketLogging",
"s3:ListBucketVersions",
"s3:ReplicateTags",
"s3:RestoreObject",
"s3:ListBucket",
"s3:GetAccelerateConfiguration",
"s3:GetBucketPolicy",
"s3:PutEncryptionConfiguration",
"s3:GetEncryptionConfiguration",
"s3:GetObjectVersionTorrent",
"s3:AbortMultipartUpload",
"s3:PutBucketTagging",
"s3:GetBucketRequestPayment",
"s3:GetAccessPointPolicyStatus",
"s3:GetObjectTagging",
"s3:GetMetricsConfiguration",
"s3:PutBucketVersioning",
"s3:GetBucketPublicAccessBlock",
"s3:ListBucketMultipartUploads",
"s3:PutMetricsConfiguration",

```

        "s3:PutObjectVersionTagging",
        "s3:GetBucketVersioning",
        "s3:GetBucketAcl",
        "s3:PutInventoryConfiguration",
        "s3:GetObjectTorrent",
        "s3:PutBucketWebsite",
        "s3:PutBucketRequestPayment",
        "s3:PutObjectRetention",
        "s3:GetBucketCORS",
        "s3:GetBucketLocation",
        "s3:GetAccessPointPolicy",
        "s3:GetObjectVersion",
        "s3:GetAccessPoint",
        "s3:GetAccountPublicAccessBlock",
        "s3:ListAllMyBuckets",
        "s3:ListAccessPoints",
        "s3:ListJobs"
    ],
    "Resource": [
        "arn:aws:s3:::<YOUR_BUCKET_NAME>",
        "arn:aws:s3:::<YOUR_BUCKET_NAME>/*"
    ]
}
]
}

```

4. Attach policy to user

```

aws iam put-user-policy \
  --user-name circleci-server \
  --policy-name circleci-server \
  --policy-document file://policy.json

```

5. Create Access Key for user circleci-server



You will need this when you configure your server installation later.

```

aws iam create-access-key --user-name circleci-server

```

The result should look like this:

```

{
  "AccessKey": {
    "UserName": "circleci-server",
    "Status": "Active",
    "CreateDate": "2017-07-31T22:24:41.576Z",
    "SecretAccessKey": <AWS_SECRET_ACCESS_KEY>,
  }
}

```

```
}  
  "AccessKeyId": <AWS_ACCESS_KEY_ID>  
}
```

CircleCI Server v3.x Installation Phase 2

Before you begin with the CircleCI server v3.x core services installation phase, ensure all [prerequisites](#) are met.

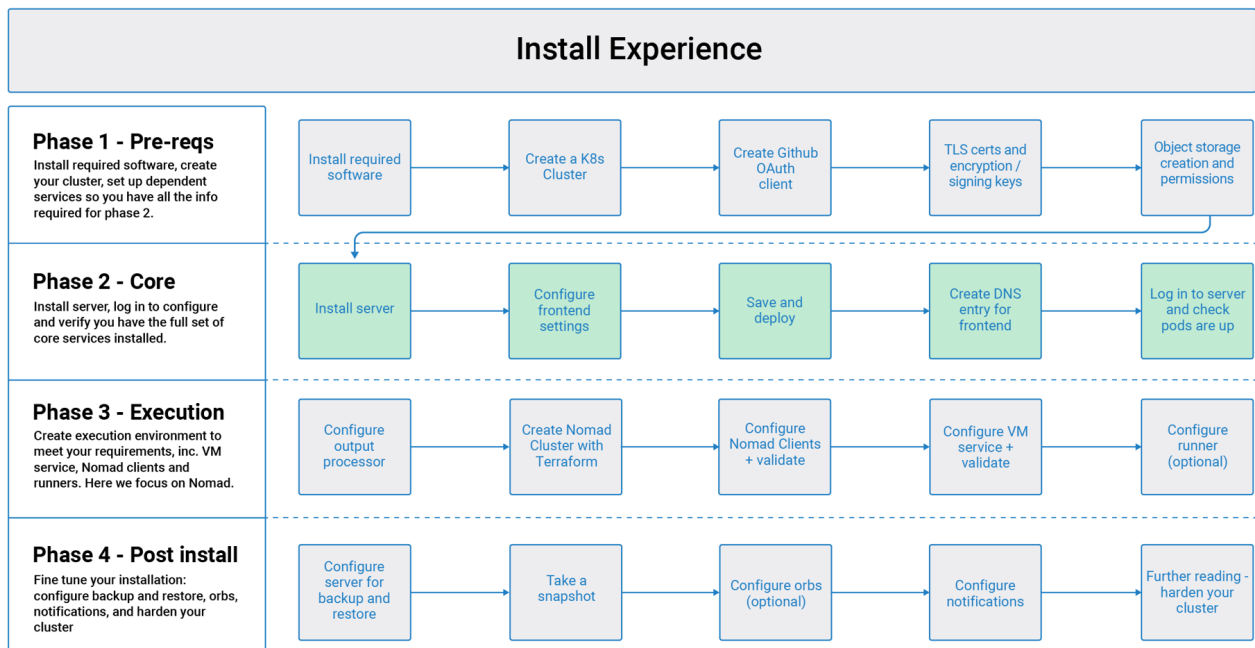


Figure 4. Installation Experience Flow Chart Phase 2



In the following sections replace any items or credentials displayed between `< >` with your details.

Phase 2: Core services installation

CircleCI server v3.x uses [KOTS](#) from [Replicated](#) for installation management and distribution.

1. Ensure you are running the minimum KOTS version (1.65.0) by running the command:

```
kubectl kots version
```



The KOTS command opens up a tunnel to the admin console. If running on Windows inside WSL2, the port is not available on the host machine. Turning WSL off and back on should resolve the issue. For more information, please see <https://github.com/microsoft/WSL/issues/4199>.

2. From the terminal, run (if you are installing behind a proxy see [Installing behind HTTP Proxy](#)):

```
kubectl kots install circleci-server
```

You will be prompted for:

- namespace for the deployment
 - password for the KOTS Admin Console
3. When complete, you should be provided with a URL to access the KOTS admin console, usually `http://localhost:8800`.



If you need to get back to the KOTS admin console at a later date, run: `kubectl kots admin-console -n <YOUR_CIRCLECI_NAMESPACE>`



Once you have created your namespace, we recommend setting your `kubectl` context too with the following command: `kubectl config set-context --current --namespace <namespace>`

Installing behind an HTTP Proxy (optional)

If you wish to install CircleCI server behind a proxy, the following command structure should be used for step 2 above (for more information see the KOTS docs [here](#)):

```
kubectl kots install circleci-server --http-proxy <YOUR_HTTP_PROXY_URI> --https-proxy <https-proxy> --no-proxy <YOUR_NO_PROXY_LIST>
```

The load balancer endpoints must be added to the no-proxy list for the following services: `output processor` and `vm-service`. This is because the no-proxy list is shared between the application and build-agent. The application and build-agent are assumed to be behind the same firewall and therefore cannot have a proxy between them.

For further information see the [Configuring a Proxy](#) guide.

Frontend Settings

Frontend settings control the web-application-specific aspects of the CircleCI system.

Frontend Settings ^

CircleCI Domain Name

Frontend Replicas

Frontend TLS Private Key

Frontend TLS Certificate

Enable HTTPS Termination

Automatically request and manage Let's Encrypt TLS certificates

Use AWS ACM to manage TLS certificates

Private load balancers

Serve Unsafe Artifacts

Encryption v

Github v

Object Storage v

Output Processor v

Nomad v

VM Service v

Postgres v

MongoDB v

Vault v

Observability and monitoring v

Email Notifications v

HTTP Proxy v

Randomized credentials and ... v

Kong v

Frontend Settings

Please see [documentation](#) for complete details on all the configuration options available to you here.

CircleCI Domain Name Required

Domain for your server install (e.g circleci.example.com)

Frontend Replicas

Increase this number to add additional pods, to scale out frontend. See scaling documentation for more details.

1

Default value: 1

Frontend TLS Private Key

PEM encoded TLS private key for Web application. Leave this field empty and CircleCI will generate a self signed certificate for you.

Frontend TLS Certificate

PEM encoded TLS certificate for Web application. Leave this field empty and CircleCI will generate a self signed certificate for you.

☒ Enable HTTPS Termination

If this setting is enabled, CircleCI will terminate TLS. Disabling this box requires that TLS be terminated upstream of the application.

☐ Automatically request and manage Let's Encrypt TLS certificates Recommended

If enabled, Let's Encrypt TLS certificates will automatically be requested and renewed for you. It requires the CircleCI Domain Name be a publicly accessible DNS record.

☐ Use AWS ACM to manage TLS certificates Recommended

If enabled, TLS certificates will automatically be requested from AWS ACM and renewed for you.

⚠️WARNING: This will recreate frontend's service which will recreate the load balancer. If you are updating your deployed settings, then you will need to route your frontend domain to the new loadbalancer.

☐ Private load balancers

Ticking this box will make the load balancers for the frontend private, resulting in no public IPs being assigned to them. If you change this setting after the initial deployment, you might need to re-deploy the traefik service for it to take effect. Note that this only works for GKE and EKS installations.

☐ Serve Unsafe Artifacts

Warning! Enabling unsafe content-types allows specially-crafted artifacts to gain control of users' CircleCI accounts.

Figure 5. Frontend Settings

Complete the fields described below.

- **CircleCI Domain Name (required)** - Enter the domain name you specified when creating your Frontend TLS key and certificate.
- **Frontend Replicas** - Used to increase the amount of traffic that can be handled by the frontend.
- **Frontend TLS Private Key (required)** - You created this during your prerequisite steps. You can retrieve this value with the following command:

```
cat /etc/letsencrypt/live/<CIRCLECI_SERVER_DOMAIN>/privkey.pem
```

- **Frontend TLS Certificate (required)** - You created this during your prerequisite steps. You can retrieve this value with the following command:

```
cat /etc/letsencrypt/live/<CIRCLECI_SERVER_DOMAIN>/fullchain.pem
```

- **Private Load Balancer (optional)** - Load balancer does not generate external IP addresses.



If you are selecting the option to use private load balancers, the Let's Encrypt option will no longer work and become unavailable.

For the **Frontend TLS private key and certificate** you have 4 options:

- You can supply a private key and certificate
- Check the box that allows [Let's Encrypt](#) to automatically request and manage certificates for you.
- Check the box that allows [AWS Certificate Manager \(ACM\)](#) to automatically request and manage certificates for you. For more information about using ACM see the [Using ACM TLS certificates](#) section below.
- You can also disable TLS termination at this point, but the system will still need to be accessed over HTTPS.

Using ACM TLS certificates

If you would like to use [AWS Certificate Manager \(ACM\)](#) to manage your TLS certificates, follow the [ACM documentation](#) for instructions on how to generate ACM certificates.

Once you have generated your certificates, enable ACM from the KOTS admin console under the Frontend section. Check the ACM box and provide your ACM ARN (Amazon Resource Name).



If you have already deployed CircleCI server, enabling ACM is a destructive change to your frontend service. The service will have to be regenerated to allow the use of your ACM certificates and so the associated loadbalancer will also be regenerated. You will need to reroute your DNS records to the new loadbalancer once you have redeployed CircleCI server.

Encryption

Encryption and artifact signing keys were created during prerequisites phase. You can enter them here now.

Frontend Settings ^

- CircleCI Domain Name
- Frontend Replicas
- Frontend TLS Private Key
- Frontend TLS Certificate
- Enable HTTPS Termination
- Use AWS ACM to manage TLS certificates
- ARN for requesting ACM certificates
- Private load balancers
- Serve Unsafe Artifacts

Encryption

Encryption and signing keys are used to secure and prevent tampering with build artifacts. See each field for instructions on how to generate these.

⚠ WARNING: There is no support for rotating these keys. If they are lost your artifacts will not be retrievable. Be sure to store a backup in a secure location.

Artifact Signing Key Required

Signing key for build artifacts. To create a new one, run `docker run circleci/server-keysets generate signing`.

.....

Artifact Encryption Key Required

Encryption key for build artifacts. To create a new one, run `docker run circleci/server-keysets generate encryption`.

.....

Figure 6. Encryption Settings

Complete the following fields:

- **Artifact Signing Key (required)**
- **Encryption Signing Key (required)**

GitHub

You created your GitHub OAuth application in the prerequisite phase. Use the data to complete the following settings:

Frontend Settings ^

- CircleCI Domain Name
- Frontend Replicas
- Frontend TLS Private Key
- Frontend TLS Certificate
- Enable HTTPS Termination
- Use AWS ACM to manage TLS certificates
- ARN for requesting ACM certificates
- Private load balancers
- Serve Unsafe Artifacts

Encryption ▾

GitHub ▾

Object Storage ▾

Output Processor ▾

Github

These settings control authorization to server using Github OAuth and allow server to update Github with build status information.

Github type

☒ Github cloud (github.com) ☐ Github Enterprise

OAuth Client ID Required

Can be created by navigating to `User settings > Developer settings > OAuth Apps > New OAuth App`. Be sure to set the Homepage URL to `https://domain_name` and the Authorization callback URL to `https://domain_name/auth/github` where `domain_name` is the domain name you set for this server install.

.....

OAuth Client Secret Required

On your OAuth application, create one by selecting `Generate a new client secret`.

.....

☒ **Disable Webhook SSL Verification**

Should only be checked if your Github installation does not trust the certificate authority that signed your CircleCI server certificates (e.g they were private-ca or self signed certificate).

Figure 7. GitHub Settings

- **GitHub Type (required)** - Select Cloud or Enterprise (on premises).
- **OAuth Client ID (required)** - The OAuth Client ID provided by GitHub.
- **OAuth Client Secret (required)** - The OAuth Client Secret provided by GitHub.
- **GitHub Enterprise Fingerprint** - Required when using a proxy. Include the output of `ssh-keyscan github.example.com` in the text field.

Object storage

You created your Object Storage Bucket and Keys in the prerequisite steps. Use the data to complete the following settings depending on your platform.

Frontend Settings ^

- CircleCI Domain Name
- Frontend Replicas
- Frontend TLS Private Key
- Frontend TLS Certificate
- Enable HTTPS Termination
- Use AWS ACM to manage TLS certificates
- ARN for requesting ACM certificates
- Private load balancers
- Serve Unsafe Artifacts

Object Storage

Server hosts build artifacts, test results and other state in object storage. Configure your object storage location and access credentials here.

Storage Bucket Name Required

rohara-devv

Storage Object Expiry Required

Number of days after which objects in your object storage expire. This should match your object storage's retention policy. Set to 0 to disable.

0

Default value: 0

Object Storage type

☒ S3 ☐ Google Cloud Storage

AWS S3 Region

If using AWS S3, this is the region your S3 bucket lives in (e.g. [us-east-1](#)). Ignored if S3 endpoint is also supplied.

us-west-2

IAM Access Type

☒ IAM Keys ☐ IAM role

Access Key ID

Access key for S3 bucket access

.....

Secret Key

Secret key for S3 bucket access

.....

Figure 8. Object Storage Settings

S3 compatible

- **Storage Bucket Name (required)** - The bucket used for server.
- **AWS S3 Region (optional)** - AWS region of bucket if your provider is AWS. S3 Endpoint is ignored if this option is set.
- **S3 Endpoint (optional)** - API endpoint of S3 storage provider. Required if your provider is not AWS. AWS S3 Region is ignored if this option is set.
- **Storage Object Expiry (required)** - Number of days to retain your test results and artifacts. Set to 0 to disable and retain objects indefinitely.

Authentication

One of the following is required. Either select IAM keys and provide:

- **Access Key ID (required)** - Access Key ID for S3 bucket access.
- **Secret Key (required)** - Secret Key for S3 bucket access.

- **policy.json (required)** - As outlined in the [server AWS Installation Guide](#), on page 11.

Or select IAM role and provide:

- **Role ARN (required)** - [Role ARN for Service Accounts](#) (Amazon Resource Name) for S3 bucket access.
- **policy.json (required)** - As outlined in the [server AWS Installation Guide](#), on page 8.

Skip over the next few sections - **Output Processor**, **Nomad** and **VM Service**. We will set these up in the next phase of the installation.

Postgres, MongoDB, Vault settings

You can skip these sections unless you plan on using an existing Postgres, MongoDB or Vault instance, in which case, see the [Externalizing Services doc](#). By default, CircleCI server v3.x will create its own Postgres, MongoDB and Vault instances within the CircleCI namespace. The instances inside the CircleCI namespace will be included in the CircleCI backup and restore process.

Save and deploy

Once you have completed the fields detailed above, you can deploy. The deployment installs the core services and provides you with an IP address for the Kong load balancer. That IP address is critical in setting up a DNS record and completing the first phase of the installation.



From server v3.3.0, we have replaced [Traefik](#) with [Kong](#) as our reverse proxy. However, to minimize disruption when upgrading, we chose not to rename the service used by Kong. Although you will see a service named `circleci-server-traefik`, this service is actually for Kong.

Create DNS entry

Create a DNS entry for your Kong load balancer, for example, `circleci.your.domain.com` and `app.circleci.your.domain.com`. The DNS entry should align with the DNS names used when creating your TLS certificate and GitHub OAuth app during the prerequisites steps. All traffic will be routed through this DNS record.

You need the IP address or, if using AWS, the DNS name of the Kong load balancer. You can find this information with the following command:

```
kubectl get service circleci-server-traefik --namespace=<YOUR_CIRCLECI_NAMESPACE>
```

For more information on adding a new DNS record, see the following documentation:

- [Managing Records](#) (GCP)
- [Creating records by using the Amazon Route 53 Console](#) (AWS)



The Kong load balancer has a healthcheck that serves a JSON payload at <https://loadbalancer-address/status>

Validation

You should now be able to navigate to your CircleCI server installation and log in to the application successfully.

Now we will move on to build services. It may take a while for all your services to be up. You can periodically check by running the following command (you are looking for the “frontend” pod to show a status of *running* and **ready** should show 1/1):

```
kubectl get pods -n <YOUR_CIRCLECI_NAMESPACE>
```

CircleCI Server v3.x Installation Phase 3

Before you begin with the CircleCI server v3.x execution installation phase, ensure you have run through [Phase 1 – Prerequisites](#) and [Phase 2 - Core services installation](#).

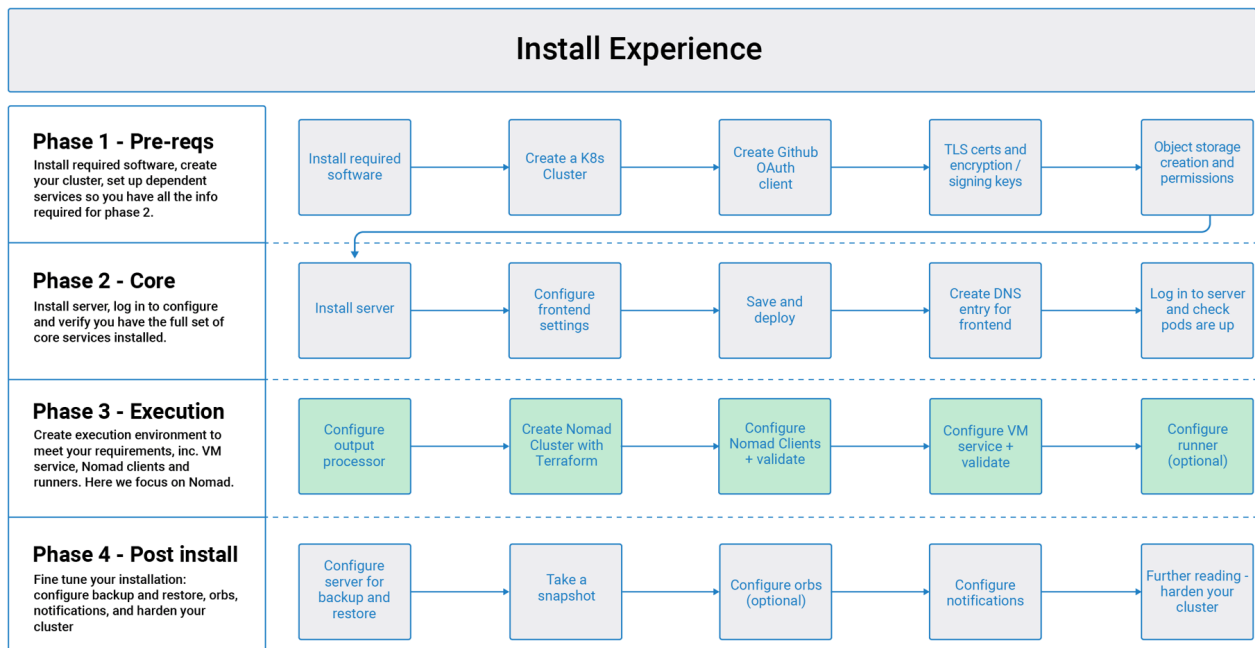


Figure 9. Installation Experience Flow Chart Phase 3



In the following sections, replace any items or credentials displayed between `< >` with your details.

Phase 3: Execution environment installation

Output Processor

Output processor is responsible for handling the output from Nomad clients. It is a key service to scale if you find your system slowing down. We recommend increasing the output processor replica set to scale the service up to meet demand.

Access the KOTS Admin Console by running the following command, substituting your namespace:
`kubectl kots admin-console -n <YOUR_CIRCLECI_NAMESPACE>`

Locate and enter the following in Settings:

1. **Output Processor Load Balancer Hostname** - The following command provides the IP address of the service:

```
kubectl get service output-processor --namespace=<YOUR_CIRCLECI_NAMESPACE>
```

2. **Save your configuration.** You will deploy and validate your setup after you complete Nomad

client setup.

Nomad Clients

As mentioned in the [Overview](#), Nomad is a workload orchestration tool that CircleCI uses to schedule (through Nomad Server) and run (through Nomad Clients) CircleCI jobs.

Nomad clients are installed outside of the Kubernetes cluster, while their control plane (Nomad Server) is installed within the cluster. Communication between your Nomad Clients and the Nomad control plane is secured with mTLS. The mTLS certificate, private key, and certificate authority will be output after you complete installation of the Nomad Clients.

Once completed, you can update your CircleCI server configuration so your Nomad control plane can communicate with your Nomad Clients.

Cluster Creation with Terraform

CircleCI curates Terraform modules to help install Nomad clients in your chosen cloud provider. You can browse the modules in our [public repository](#), including example Terraform config files (`main.tf`) for both AWS and GCP. Some information about your cluster and server installation is required to complete your `main.tf`. How to get this information is described in the following sections.



If you would also like to set up Nomad Autoscaler at this stage, see the [Nomad Autoscaler](#) section of this guide, as some of the requirements can be included in this Terraform setup.

AWS

You need some information about your cluster and server installation to complete the required fields for the Terraform configuration file (`main.tf`). A full example, as well as a full list of variables, can be found [here](#).

- **Server_endpoint** - You need to know the Nomad Server endpoint, which is the external IP address of the nomad-server-external Loadbalancer. You can get this information with the following command:

```
kubectl get service nomad-server-external --namespace=<YOUR_CIRCLECI_NAMESPACE>
```

- **Subnet ID (subnet)**, **VPC ID (vpcId)**, and **DNS server (dns_server)** of your cluster. Run the following command to get the cluster VPC ID (vpcId), CIDR block (serviceIpv4Cidr), and subnets (subnetIds):

```
aws eks describe-cluster --name=<YOUR_CLUSTER_NAME>
```

This returns something similar to the following:

```
{...
  "resourcesVpcConfig": {
    "subnetIds": [
      "subnet-033a9fb4be69",
      "subnet-04e89f9eef89",
      "subnet-02907d9f35dd",
      "subnet-0fbc63006c5f",
      "subnet-0d683b6f6ba8",
      "subnet-079d0ca04301"
    ],
    "clusterSecurityGroupId": "sg-022c1b544e574",
    "vpcId": "vpc-02fdfff4c",
    "endpointPublicAccess": true,
    "endpointPrivateAccess": false
  },
  "kubernetesNetworkConfig": {
    "serviceIpv4Cidr": "10.100.0.0/16"
  },
}
```

Then, using the VPCID you just found, run the following command to get the CIDR Block for your cluster. For AWS, the DNS Server is the third IP in your CIDR block (**CidrBlock**), for example your CIDR block might be **10.100.0.0/16**, so the third IP would be **10.100.0.2**.

```
aws ec2 describe-vpcs --filters Name=vpc-id,Values=<YOUR_VPCID>
```

This returns something like the following:

```
{...
  "CidrBlock": "192.168.0.0/16",
  "DhcpOptionsId": "dopt-9cff",
  "State": "available",
  "VpcId": "vpc-02fdfff4c"
}
```

Once you have filled in the appropriate information, you can deploy your Nomad clients by running the following command from within the directory of the **main.tf** file:

```
terraform init
```

```
terraform plan
```

```
terraform apply
```

After Terraform is done spinning up the Nomad client(s), it outputs the certificates and keys needed for configuring the Nomad control plane in CircleCI server. Copy them somewhere safe. The apply process usually only takes a minute.

Nomad Autoscaler

Nomad provides a utility to automatically scale up or down your Nomad clients, provided your clients are managed by a cloud provider's autoscaling resource. With Nomad Autoscaler, you only need to provide permission for the utility to manage your autoscaling resource and specify where it is located. You can enable this resource via KOTS, which deploys the Nomad Autoscaler service along with your Nomad servers. Below, we go through how to set up Nomad Autoscaler for your provider.



The maximum and minimum Nomad client count overwrite the corresponding values set when you created your autoscaling group or managed instance group. It is recommended that you keep these values and those used in your Terraform the same so that they do not compete.

If you do not require this service, click the **Save config** button to update your installation and redeploy server.

AWS

1. Create an IAM user or role and policy for Nomad Autoscaler. You may take one of the following approaches:
 - Our [nomad module](#) creates an IAM user and outputs the keys if you set variable `nomad_auto_scaler = true`. You may reference the example in the link for more details. If you have already created the clients, you can update the variable and run `terraform apply`. The created user's access key and secret will be available in Terraform's output.
 - You may also create a Nomad Autoscaler IAM user manually with the IAM policy below. Then you need to generate an access and secret key for this user.
 - You may create a [Role for Service Accounts](#) for Nomad Autoscaler and attach the following IAM policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": [
        "autoscaling:CreateOrUpdateTags",
        "autoscaling:UpdateAutoScalingGroup",
        "autoscaling:TerminateInstanceInAutoScalingGroup"
      ]
    }
  ]
}
```

```

    ],
    "Resource": "<<Your Autoscaling Group ARN>>"
  },
  {
    "Sid": "VisualEditor1",
    "Effect": "Allow",
    "Action": [
      "autoscaling:DescribeScalingActivities",
      "autoscaling:DescribeAutoScalingGroups"
    ],
    "Resource": "*"
  }
]
}

```

2. In your KOTS Admin Console, set Nomad Autoscaler to **enabled**.
3. Set Max Node Count* - This overwrites what is currently set as the max for your ASG. It is recommended to keep this value and what was set in your Terraform the same.
4. Set Min Node Count* - This overwrites what is currently set as the min for your ASG. It is recommended to keep this value and what was set in your Terraform as the same.
5. Select cloud provider: **AWS EC2**.
6. Add the region of the autoscaling group.
7. You can chose one of the following:
 - a. Add the Nomad Autoscaler user's access key and secret key.
 - b. Or, the Nomad Autoscaler role's ARN.
8. Add the name of the autoscaling Group your Nomad clients were created in.

Configure and Deploy

Now that you have successfully deployed your Nomad clients, you can configure CircleCI server and the Nomad control plane. Access the KOTS Admin Console by running the following command, substituting your namespace: **kubectl kots admin-console -n <YOUR_CIRCLECI_NAMESPACE>**

Enter the following in Settings:

- **Nomad Load Balancer (required)**

```
kubectl get service nomad-server-external --namespace=<YOUR_CIRCLECI_NAMESPACE>
```

- **Nomad Server Certificate (required)** - Provided in the output from **terraform apply**
- **Nomad Server Private Key (required)** - Provided in the output from **terraform apply**
- **Nomad Server Certificate Authority (CA) Certificate (required)** - Provided in the output from **terraform apply**
- **Build Agent Image** - If you want to use a custom Docker registry to supply the CircleCI Build

Agent, contact customer support for assistance.

Click the **Save config** button to update your installation and redeploy server.

Nomad Clients Validation

CircleCI has created a project called [realitycheck](#) which allows you to test your Server installation. We are going to follow the project so we can verify that the system is working as expected. As you continue through the next phase, sections of realitycheck will move from red to green.

To run realitycheck, you need to clone the repository. Depending on your GitHub setup, you can use one of the following commands:

GitHub Cloud

```
git clone -b server-3.0 https://github.com/circleci/realitycheck.git
```

GitHub Enterprise

```
git clone -b server-3.0 https://github.com/circleci/realitycheck.git
git remote set-url origin <YOUR_GH_REPO_URL>
git push
```

Once you have successfully cloned the repository, you can follow it from within your CircleCI server installation. You need to set the following variables. For full instructions please see the [repository readme](#).

Table 1. Environmental Variables

Name	Value
CIRCLE_HOSTNAME	<YOUR_CIRCLECI_INSTALLATION_URL>
CIRCLE_TOKEN	<YOUR_CIRCLECI_API_TOKEN>

Table 2. Contexts

Name	Environmental Variable Key	Environmental Variable Value
org-global	CONTEXT_END_TO_END_TEST_VAR	Leave blank
individual-local	MULTI_CONTEXT_END_TO_END_VAR	Leave blank

Once you have configured the environmental variables and contexts, rerun the realitycheck tests. You should see the features and resource jobs complete successfully. Your test results should look something like the following:

Pipeline	Status	Workflow	Branch / Commit	Start	Duration	Actions
realitycheck 3	Running	vm_jobs	master 368b13d	19s ago	1m 48s	...
Jobs		remote_docker 60			1m 19s	
		machine_dlc 62			1m 29s	
		machine 63			1m 23s	
		docker_layer_caching 61			1m 19s	
	Success	resource_class_jobs	master 368b13d	29s ago	29s	...
Jobs		xlarge 55			20s	
		small 58			2s	
		medium+ 57			20s	
		medium 59			3s	
		large 56			21s	
	Success	feature_jobs	master 368b13d	2m ago	31s	...
Jobs		write_workspace 51			6s	
		save_and_restore_cache 49			27s	
		read_workspace 54			2s	
		multi-contexts 52			6s	
		contexts 53			5s	
		artifacts_test_results 50			7s	

VM service

VM service configures VM and remote docker jobs. You can configure a number of options for VM service, such as scaling rules. VM service is unique to EKS and GKE installations because it specifically relies on features of these cloud providers.

AWS

1. Get the Information Needed to Create Security Groups

The following command returns your VPC ID (`vpcId`), CIDR Block (`serviceIpv4Cidr`), Cluster Security Group ID (`clusterSecurityGroupId`) and Cluster ARN (`arn`) values, which you need throughout this section:

```
aws eks describe-cluster --name=<your-cluster-name>
```

2. Create a security group

Run the following commands to create a security group for VM service:

```
aws ec2 create-security-group --vpc-id "<YOUR_VPCID>" --description "CircleCI VM Service security group" --group-name "circleci-vm-service-sg"
```

This outputs a GroupID to be used in the next steps:

```
{
  "GroupId": "sg-0cd93e7b30608b4fc"
```

```
}
```

3. Apply security group Nomad

Use the security group you just created and CIDR block values to apply the security group to the following:

```
aws ec2 authorize-security-group-ingress --group-id "<YOUR_GroupId>" --protocol tcp --port 22 --cidr "<YOUR_serviceIpv4Cidr>"
```

```
aws ec2 authorize-security-group-ingress --group-id "<YOUR_GroupId>" --protocol tcp --port 2376 --cidr "<YOUR_serviceIpv4Cidr>"
```



If you created your Nomad Clients in a different subnet from CircleCI server, you need to rerun the above two commands with each subnet CIDR.

4. Apply the security group for SSH

Run the following command to apply the security group rules so users can SSH into their jobs:

```
aws ec2 authorize-security-group-ingress --group-id "<YOUR_GroupId>" --protocol tcp --port 54782
```

5. Create user

Create a new user with programmatic access:

```
aws iam create-user --user-name circleci-vm-service
```



Optionally, vm-service does support the use of a [service account role](#) in place of AWS keys. If you would prefer to use a role, follow these [instructions](#) using the policy in step 6 below. **When creating the service account, please ensure the name is set as `vm-service`.**

Once done, you may skip to step 9 which is enabling vm-service in KOTS.

6. Create policy

Create a `policy.json` file with the following content. You should fill in the ID of the VM Service security group created in step 2 (`VMServiceSecurityGroupId`) and VPC ID (`vpcID`) below.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Action": "ec2:RunInstances",
```

```

    "Effect": "Allow",
    "Resource": [
        "arn:aws:ec2:*:*:image/*",
        "arn:aws:ec2:*:*:snapshot/*",
        "arn:aws:ec2:*:*:key-pair/*",
        "arn:aws:ec2:*:*:launch-template/*",
        "arn:aws:ec2:*:*:network-interface/*",
        "arn:aws:ec2:*:*:placement-group/*",
        "arn:aws:ec2:*:*:volume/*",
        "arn:aws:ec2:*:*:subnet/*",
        "arn:aws:ec2:*:*:security-group/<YOUR_VMSecurityGroupID>"
    ]
},
{
    "Action": "ec2:RunInstances",
    "Effect": "Allow",
    "Resource": "arn:aws:ec2:*:*:instance/*",
    "Condition": {
        "StringEquals": {
            "aws:RequestTag/ManagedBy": "circleci-vm-service"
        }
    }
},
{
    "Action": [
        "ec2:CreateVolume"
    ],
    "Effect": "Allow",
    "Resource": [
        "arn:aws:ec2:*:*:volume/*"
    ],
    "Condition": {
        "StringEquals": {
            "aws:RequestTag/ManagedBy": "circleci-vm-service"
        }
    }
},
{
    "Action": [
        "ec2:Describe*"
    ],
    "Effect": "Allow",
    "Resource": "*"
},
{
    "Effect": "Allow",
    "Action": [
        "ec2:CreateTags"
    ],
    "Resource": "arn:aws:ec2:*:*:*/*",
    "Condition": {

```

```

        "StringEquals": {
            "ec2:CreateAction" : "CreateVolume"
        }
    },
    {
        "Effect": "Allow",
        "Action": [
            "ec2:CreateTags"
        ],
        "Resource": "arn:aws:ec2:*:*:*/*",
        "Condition": {
            "StringEquals": {
                "ec2:CreateAction" : "RunInstances"
            }
        }
    },
    {
        "Action": [
            "ec2:CreateTags",
            "ec2:StartInstances",
            "ec2:StopInstances",
            "ec2:TerminateInstances",
            "ec2:AttachVolume",
            "ec2:DetachVolume",
            "ec2>DeleteVolume"
        ],
        "Effect": "Allow",
        "Resource": "arn:aws:ec2:*:*:*/*",
        "Condition": {
            "StringEquals": {
                "ec2:ResourceTag/ManagedBy": "circleci-vm-service"
            }
        }
    },
    {
        "Action": [
            "ec2:RunInstances",
            "ec2:StartInstances",
            "ec2:StopInstances",
            "ec2:TerminateInstances"
        ],
        "Effect": "Allow",
        "Resource": "arn:aws:ec2:*:*:subnet/*",
        "Condition": {
            "StringEquals": {
                "ec2:Vpc": "<YOUR_vpcID>"
            }
        }
    }
}
]

```

```
}
```

7. Attach policy to user

Once you have created the policy.json file, attach it to an IAM policy and created user.

```
aws iam put-user-policy --user-name circleci-vm-service --policy-name circleci-vm-service --policy-document file://policy.json
```

8. Create an access key and secret for the user

If you have not already created them, you will need an access key and secret for the `circleci-vm-service` user. You can create those by running the following command:

```
aws iam create-access-key --user-name circleci-vm-service
```

9. Configure server

Configure VM Service through the KOTS Admin Console. Details of the available configuration options can be found in the [VM Service](#) guide.

Once you have configured the fields, **save your config** and deploy your updated application.

VM Service Validation

Once you have configured and deployed CircleCI server, you should validate that VM Service is operational. You can rerun the [realitycheck](#) project within your CircleCI installation and you should see the VM Service Jobs complete. At this point, all tests should pass.

Runner

Overview

CircleCI runner does not require any additional server configuration. Server ships ready to work with runner. However, you need to create a runner and configure the runner agent to be aware of your server installation. For complete instructions for setting up runner, see the [runner documentation](#).



Runner requires a namespace per organization. Server can have many organizations. If your company has multiple organizations within your CircleCI installation, you need to set up a runner namespace for each organization within your server installation.

CircleCI Server v3.x Installation Phase 4

Before you begin with the CircleCI server v3.x post installation phase, ensure you have run through [Phase 1 – Prerequisites](#), [Phase 2 - Core services installation](#) and [Phase 3 - Build services installation](#).

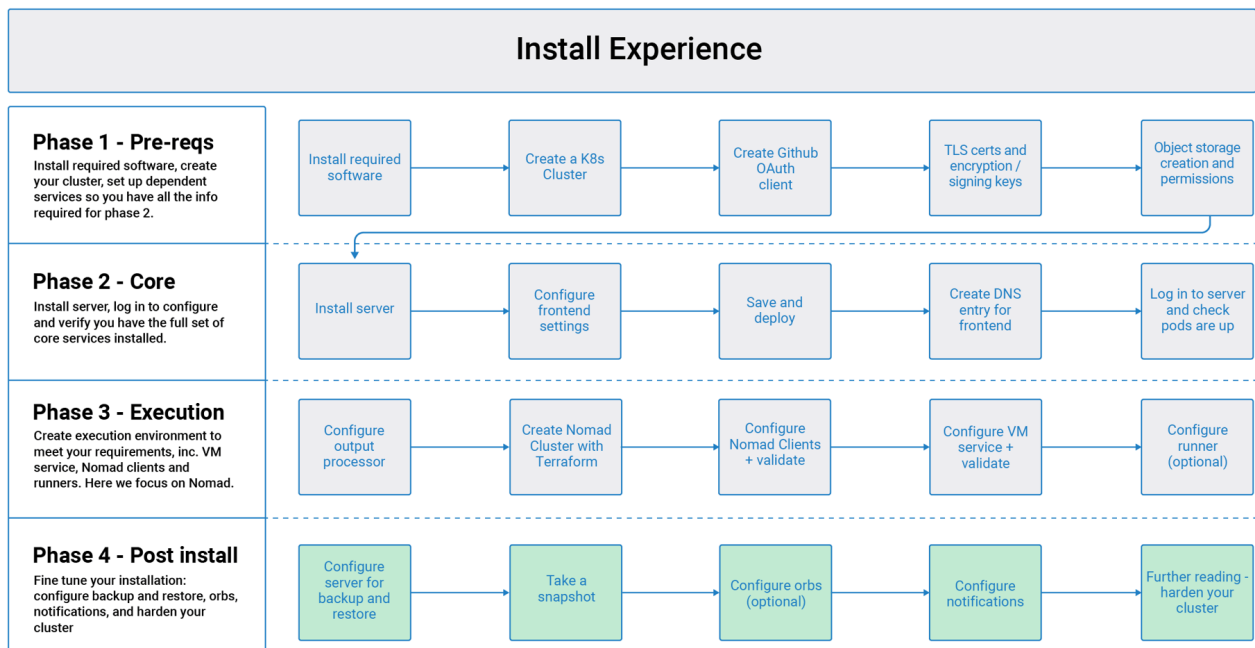


Figure 10. Installation Experience Flow Chart Phase 4



In the following sections, replace any items or credentials displayed between `< >` with your details.

Phase 4: Post installation

Set up backup and restore

Backups of CircleCI server can be created through [KOTS](#). To enable backup support, you will need to install and configure [Velero](#) on your cluster. Velero was listed in the installation prerequisites section, so you should already have it.

Server 3.x backups on AWS

These instructions were sourced from the Velero documentation [here](#).

Step 1 - Create an AWS S3 bucket

```
BUCKET=<YOUR_BUCKET>
REGION=<YOUR_REGION>
aws s3api create-bucket \
  --bucket $BUCKET \
  --region $REGION \
```

```
--create-bucket-configuration LocationConstraint=$REGION
```



`us-east-1` does not support a `LocationConstraint`. If your region is `us-east-1`, omit the bucket configuration.

Step 2 - Set up permissions for Velero

- Create an IAM user

```
aws iam create-user --user-name velero
```

- Attach policies to give user `velero` the necessary permissions:

```
cat > velero-policy.json <<EOF
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "ec2:DescribeVolumes",
        "ec2:DescribeSnapshots",
        "ec2:CreateTags",
        "ec2:CreateVolume",
        "ec2:CreateSnapshot",
        "ec2>DeleteSnapshot"
      ],
      "Resource": "*"
    },
    {
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:DeleteObject",
        "s3:PutObject",
        "s3:AbortMultipartUpload",
        "s3:ListMultipartUploadParts"
      ],
      "Resource": [
        "arn:aws:s3:::${BUCKET}/*"
      ]
    },
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket"
      ],
      "Resource": [
```

```

        "arn:aws:s3:::${BUCKET}"
    ]
}
EOF

```

```

aws iam put-user-policy \
  --user-name velero \
  --policy-name velero \
  --policy-document file://velero-policy.json

```

- Create an access key for user **velero**

```

aws iam create-access-key --user-name velero

```

The result should look like this:

```

{
  "AccessKey": {
    "UserName": "velero",
    "Status": "Active",
    "CreateDate": "2017-07-31T22:24:41.576Z",
    "SecretAccessKey": <AWS_SECRET_ACCESS_KEY>,
    "AccessKeyId": <AWS_ACCESS_KEY_ID>
  }
}

```

- Create a Velero-specific credentials file (for example: `./credentials-velero`) in your local directory, with the following contents:

```

[default]
aws_access_key_id=<AWS_ACCESS_KEY_ID>
aws_secret_access_key=<AWS_SECRET_ACCESS_KEY>

```

where the `AWS_ACCESS_KEY_ID` and `AWS_SECRET_ACCESS_KEY` placeholders are values returned from the `create-access-key` request in the previous step.

Step 3 - Install and start Velero

- Run the following `velero install` command. This creates a namespace called **velero** and installs all the necessary resources to run Velero. Make sure that you pass the correct file name containing the AWS credentials that you created in [Step 2](#).



KOTS backups require [restic](#) to operate. When installing Velero, ensure that you

have the `--use-restic` flag set, as shown below:

```
velero install \
  --provider aws \
  --plugins velero/velero-plugin-for-aws:v1.2.0 \
  --bucket $BUCKET \
  --backup-location-config region=$REGION \
  --snapshot-location-config region=$REGION \
  --secret-file ./credentials-velero \
  --use-restic \
  --wait
```

- Once Velero is installed on your cluster, check the new `velero` namespace. You should have a Velero deployment and a restic daemonset, for example:

```
$ kubectl get pods --namespace velero
```

NAME	READY	STATUS	RESTARTS	AGE
restic-5vlww	1/1	Running	0	2m
restic-94ptv	1/1	Running	0	2m
restic-ch6m9	1/1	Running	0	2m
restic-mknws	1/1	Running	0	2m
velero-68788b675c-dm2s7	1/1	Running	0	2m

As restic is a daemonset, there should be one pod for each node in your Kubernetes cluster.

Server 3.x backups with S3 Compatible Storage

The following steps assume you are using S3-compatible object storage, but not necessarily AWS S3, for your backups. It is also assumed you have met the [prerequisites](#).

These instructions were sourced from the Velero documentation [here](#).

Step 1 - Configure `mc` client

To start, configure `mc` to connect to your storage provider:

```
# Alias can be any name as long as you use the same value in subsequent commands
export ALIAS=my-provider
mc alias set $ALIAS <YOUR_MINIO_ENDPOINT> <YOUR_MINIO_ACCESS_KEY_ID> <YOUR_MINIO_SECRET_ACCESS_KEY>
```

You can verify your client is correctly configured by running `mc ls my-provider` and you should see the buckets in your provider enumerated in the output.

Step 2 - Create a bucket

Create a bucket for your backups. It is important that a new bucket is used, as Velero cannot use a preexisting bucket that contains other content.

```
mc mb ${ALIAS}/<YOUR_BUCKET>
```

Step 3 - Create a user and policy

Next, create a user and policy for Velero to access your bucket.



In the following snippet `<YOUR_MINIO_ACCESS_KEY_ID>` and `<YOUR_MINIO_SECRET_ACCESS_KEY>` refer to the credentials used by Velero to access MinIO.

```
# Create user
mc admin user add $ALIAS <YOUR_MINIO_ACCESS_KEY_ID> <YOUR_MINIO_SECRET_ACCESS_KEY>

# Create policy
cat > velero-policy.json << EOF
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:*"
      ],
      "Resource": [
        "arn:aws:s3:::<YOUR_BUCKET>",
        "arn:aws:s3:::<YOUR_BUCKET>/*"
      ]
    }
  ]
}
EOF

mc admin policy add $ALIAS velero-policy velero-policy.json

# Bind user to policy
mc admin policy set $ALIAS velero-policy user=<YOUR_VELERO_ACCESS_KEY_ID>
```

Finally, you add your new user's credentials to a file (`./credentials-velero` in this example) with the following contents:

```
[default]
aws_access_key_id=<YOUR_VELERO_ACCESS_KEY_ID>
aws_secret_access_key=<YOUR_VELERO_SECRET_ACCESS_KEY>
```

Step 4 - Install and start Velero

Run the following `velero install` command. This creates a namespace called `velero` and installs all the necessary resources to run Velero.



KOTS backups require `restic` to operate. When installing Velero, ensure that you have the `--use-restic` flag set, as shown below:

```
velero install --provider aws \
  --plugins velero/velero-plugin-for-aws:v1.2.0 \
  --bucket <YOUR_BUCKET> \
  --secret-file ./credentials-velero \
  --use-volume-snapshots=false \
  --use-restic \
  --backup-location-config region=minio,s3ForcePathStyle="true",s3Url=<YOUR_ENDPOINT> \
  --wait
```

Once Velero is installed on your cluster, check the new `velero` namespace. You should have a Velero deployment and a restic daemonset, for example:

```
$ kubectl get pods --namespace velero
```

NAME	READY	STATUS	RESTARTS	AGE
restic-5vlww	1/1	Running	0	2m
restic-94ptv	1/1	Running	0	2m
restic-ch6m9	1/1	Running	0	2m
restic-mknws	1/1	Running	0	2m
velero-68788b675c-dm2s7	1/1	Running	0	2m

As restic is a daemonset, there should be one pod for each node in your Kubernetes cluster.

Creating backups

Now that Velero is installed on your cluster, you should see the snapshots option in the navbar of the management console.

Dashboard GitOps Snapshots

If you see this option, you are ready to create your first backup. If you do not see this option, please refer to the [troubleshooting](#) section.

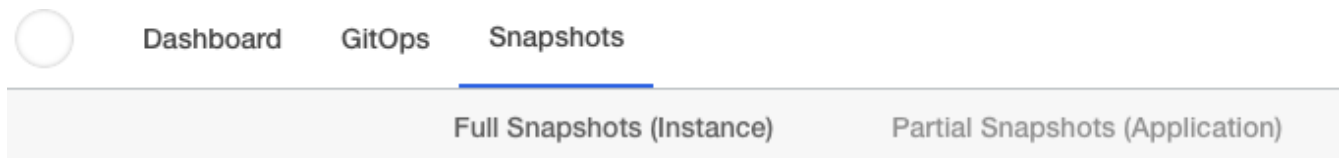
Option 1 - Create a backup with KOTS CLI

To create the backup, run the following command:

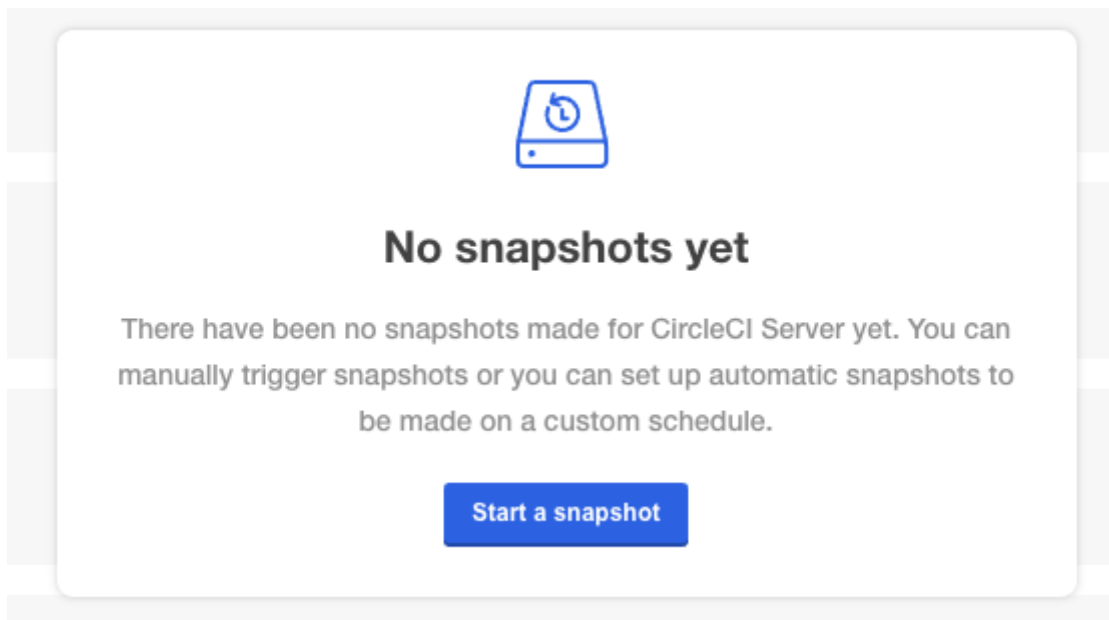
```
kubectl kots backup --namespace <your namespace>
```

Option 2 - Create a backup with KOTS Admin Console

Select **Snapshots** from the navbar. The default selection should be **Full Snapshots**, which is recommended.



Click the **Start a snapshot** button.



Orbs

Server installations include their own local orb registry. This registry is private to the server installation. All orbs referenced in project configs reference the orbs in the *server* orb registry. You are responsible for maintaining orbs. This includes:

- Copying orbs from the public registry.
- Updating orbs that may have been copied previously.
- Registering your company's private orbs, if you have any.

For more information and steps to complete these tasks, see the [Orbs on Server guide](#).

Email Notifications

Build notifications are sent by email. This section has details on how to set up build notifications by email.

Access the KOTS admin console. Get to the KOTS admin console by running the following, substituting your namespace:

```
kubectl kots admin-console -n <YOUR_CIRCLECI_NAMESPACE>
```

Locate the **Email Notifications** section in **Settings** and fill in the following details to configure email notifications for your installation.

Locate the **Email Notifications** section in **Settings** and fill in the following details to configure email notifications for your installation:

- **Email Submission server hostname (required)** - Host name of the submission server (for example, for Sendgrid use smtp.sendgrid.net).
- **Username (required)** - Username to authenticate to submission server. This is commonly the same as the user's email address.
- **Password (required)** - Password to authenticate to submission server.
- **Port (optional)** - Port of the submission server. This is usually either 25 or 587. While port 465 is also commonly used for email submission, it is often used with implicit TLS instead of StartTLS. Server only supports StartTLS for encrypted submission.



Outbound connections on port 25 are blocked on most cloud providers. Should you select this port, be aware that your notifications may fail to send.

- **Enable StartTLS** - Enabling this will encrypt mail submission.



StartTLS is used to encrypt mail by default, and you should only disable this if you can otherwise guarantee the confidentiality of traffic.

- **Email from address (required)** - The *from* address for the email.

Click the **Save config** button to update your installation and redeploy server.

CircleCI Server v3.x Migration

Migrating from 2.19.x to 3.x requires you to back up your 2.19 instance data (Mongo, Postgres, and Vault) and then restore that data in a waiting Server 3.x instance. If you experience problems, you can fall back to your 2.19 instance. Migration requires an already operating Server 3.x installation.

Depending on the size of your data stores, the migration can take anywhere from a few minutes to a few hours. We recommend using a staging environment before completing this process in a production environment. This not only allows you to gain a better understanding of the migration process, but also gives you an idea of how long the migration will take to complete.

Prerequisites

1. Your current CircleCI Server installation is 2.19.
2. You have taken a backup of the 2.19 instance. If you are using external datastores, they need to be backed up separately.
3. You have a new CircleCI Server 3.x [installation](#).
4. You have successfully run [realitycheck](#) with contexts before starting.
5. The migration script must be run from a machine with:
 - `kubectl` configured for the server 3.x instance
 - `ssh` access to the 2.19 services box

External Datastores Only

1. Backups have been taken of all external data stores.
2. Postgres has been updated to version 12.

Internal Datastore Only

1. You have taken a backup of the 2.19 instance.
2. You have successfully run [realitycheck](#) on the new server 3.x instance with contexts prior to starting.

Migration



Migrating to server v3.x will shut down your v2.19 application. Your v2.19 application will not be restarted, although you may manually restart it using the KOTS Admin Console.



Starting the migration process will cause downtime. It is recommended you schedule a maintenance window.



Running server 2.19 and server 3.x at the same time can cause issues with your

2.19 build data. Server 2.19 should NOT be restarted if server 3.x is running.

Step 1 - Clone the repository and run the migration script

The instructions below will clone the repository containing the server v2.19.x to server v3.x migration script.

The migration script will:

- Stop your v2.19.x application.
- Perform preflight checks to confirm namespace and datastores for 2.19.x.
- Create a tarball of your v2.19.x application's PostgreSQL and Mongo databases.
- Archive existing application data for Vault and CircleCI encryption/signing keys.
- Export the 2.19.x tarball to your v3.x installation. Exported data stores are stored in a directory named `circleci_export`, located relative to wherever the migration script is run from. This can be useful for debugging purposes.
- Perform preflight checks to confirm namespace and datastores for 3.x instance.
- Scale v3.x application deployments down to zero.
- Import the data from the previously exported tarball to your new v3.x instance.
- Scale v3.x application deployments up to one.



If you have externalized services, you can run `bash migrate.sh -v -p -m`. These `-v -p -m` flags will skip the migration of Vault, Postgres, and Mongo, respectively. Skipping all three will copy your keys from `/data/circle/circleci-encryption-keys` on the v2.19.x services machine, allowing you to `cat` these files and upload their contents to the 3.x configuration page.

In a terminal:

1. Run `git clone https://github.com/CircleCI-Public/server-scripts`.
2. Change into the `migrate` directory: `cd server-scripts/migrate`.
3. Run the migration script: `./migrate.sh`.
4. You will be prompted for the following information:
 - Username of your server 2.19 installation
 - Hostname of your server 2.19 installation
 - The path to your SSH key file for your server 2.19.x installation
 - Kubernetes namespace of your server 3.x installation
5. After the script has completed, the Signing and Encryption keys from the 2.19 instance will need to be added to the new 3.0 instance via the KOTS Admin Console. The keys will be located in `circleci_export/circle-data`.
6. The 3.x instance will either need to be updated to point at the same storage bucket that the 2.19 instance used, or the data needs to be copied over to a new bucket. The latter option ensures the

2.19 instance continues to work as expected, and so is the recommended approach if this migration is part of a test.



If a different hostname is being used in the 3.x environment, the GitHub webhooks will still be pointing to the hostname used in the 2.19 environment. The easiest way to update this is to click **Stop Building** and then **Set Up Project**. After doing this, the contexts and environment variables associated with the project will still be present.

Step 2 - Validate your migration to Server 3.0

Re-run [realitycheck](#) with contexts on your new server 3.x environment by pushing a fresh commit.

Step 3 - Update your team

Once you have successfully run [realitycheck](#), notify your team of the new CircleCI UI and URL, if it has changed.

Frequently Asked Questions

Where did all my job and build history go?

- All of your existing jobs and build history have been moved to the Legacy Jobs view. You can view the complete job history using one of the following methods:
 - Selecting Projects → PROJECT_NAME and selecting the **legacy jobs view** link at the bottom of the project's build history
 - Using the following URL pattern:
`https://<APP_DOMAIN>/pipelines/github/<ORG>/<PROJECT>/jobs`
 - For a specific job, append a job number to the URL:
`https://<APP_DOMAIN>/pipelines/github/<ORG>/<PROJECT>/jobs/<JOB_NUMBER>`

Why does nothing happen when I select "Start Building" on my project after migration?

- By default, a newly added project (a project that has never been followed) triggers a build automatically after it has been followed for the first time. If the project was or ever has been followed in 2.0 or 3.0, it will not be considered a new project or first build and a build will not be triggered after a follow. To trigger a build, perform an activity that triggers a GitHub webhook such as pushing up a new commit or branch.

I got an error "Error from server (NotFound):"

- The script assumes specific naming patterns for your Postgres and MongoDB. If you get this error, it may indicate a non-standard installation, a missing DB migration, or other issues. In this case, contact support with a support bundle and the output from the migration script.

Transitioning to pipelines

When migrating from a server v2.x to a v3.x installation you will have project configurations made before the introduction of pipelines. Pipelines are automatically enabled for server v3.x installations so all you need to do is change your project configurations (`.circleci/_config.yml`) to `version: 2.1` to access all the features described in the section above.

CircleCI Server v3.x Hardening Your Cluster

This section provides supplemental information on hardening your Kubernetes cluster.

Network Topology

A server installation basically runs three different type of compute instances: The Kubernetes nodes, Nomad clients, and external VMs.

It is highly recommended that you deploy these into separate subnets with distinct CIDR blocks. This will make it easier for you to control traffic between the different components of the system and isolate them from each other.

As always, the rule is to make as many of the resources as private as possible, applies. If your users will access your CircleCI server installation via VPN, there is no need to assign any public IP addresses at all, as long as you have a working NAT gateway setup. Otherwise, you will need at least one public subnet for the `circleci-server-traefik` load balancer.



From server v3.3.0, we have replaced [Traefik](#) with [Kong](#) as our reverse proxy. However, in order to minimize disruption when upgrading, we chose not to rename the service used by Kong. Although you will see a service named `circleci-server-traefik`, this service is actually for Kong.

However, in this case, it is also recommended to place Nomad clients and VMs in a public subnet to enable your users to SSH into jobs and scope access via networking rules.

Currently, custom subnetting is not supported for GCP. Custom subnetting support will be available in a future update/release.

Network Traffic

This section explains the minimum requirements for a server installation to work. Depending on your workloads, you might need to add additional rules to egress for Nomad clients and VMs. As nomenclature between cloud providers differs, you will probably need to implement these rules using firewall rules and/or security groups.

Where you see "external," this usually means all external IPv4 addresses. Depending on your particular setup, you might be able to be more specific (for example, if you are using a proxy for all external traffic).

It is assumed that you have configured the load balancers for Nomad, vm-service and output processor to be internal load balancers. This is the default.

The rules explained here are assumed to be stateful and for TCP connections only, unless stated otherwise. If you are working with stateless rules, you need to create matching ingress or egress rules for the ones listed here.

Kubernetes Load Balancers

Depending on your setup, your load balancers might be transparent (that is, they are not treated as a distinct layer in your networking topology). In this case, you can apply the rules from this section directly to the underlying destination or source of the network traffic. Refer to the documentation of your cloud provider to make sure you understand how to correctly apply networking security rules, given the type of load balancing you are using with your installation.

Ingress

If the traffic rules for your load balancers have not been created automatically, here are their respective ports:

Name	Port	Source	Purpose
*-server-traefik	80	External	User Interface & Frontend API
*-server-traefik	443	External	User Interface & Frontend API
vm-service	3000	Nomad clients	Communication with Nomad clients
nomad	4647	Nomad clients	Communication with Nomad clients
output-processor	8585	Nomad clients	Communication with Nomad clients

Egress

The only type of egress needed is TCP traffic to the Kubernetes nodes on the Kubernetes load balancer ports (30000-32767). This is not needed if your load balancers are transparent.

Common Rules for Compute Instances

These rules apply to all compute instances, but not to the load balancers.

Ingress

If you want to access your instances using SSH, you will need to open port 22 for TCP connections for the instances in question. It is recommended to scope the rule as closely as possible to allowed source IPs and/or only add such a rule when needed.

Egress

You most likely want all of your instances to access internet resources. This requires you to allow egress for UDP and TCP on port 53 to the DNS server within your VPC, as well as TCP ports 80 and 443 for HTTP and HTTPS traffic, respectively. Instances building jobs (that is, the Nomad clients and external VMs) also will likely need to pull code from your VCS using SSH (TCP port 22). SSH is also

used to communicate with external VMs, so it should be allowed for all instances with the destination of the VM subnet and your VCS, at the very least.

Kubernetes Nodes

Intra-node traffic

By default, the traffic within your Kubernetes cluster is regulated by networking policies. For most purposes, this should be sufficient to regulate the traffic between pods and there is no additional requirement to reduce traffic between Kubernetes nodes any further (it is fine to allow all traffic between Kubernetes nodes).

To make use of networking policies within your cluster, you may need to take additional steps, depending on your cloud provider and setup. Here are some resources to get you started:

- [Kubernetes Network Policy Overview](#)
- [Creating a Cluster Network Policy on Google Cloud](#)
- [Installing Calico on Amazon EKS](#)

Ingress

If you are using a managed service, you can check the rules created for the traffic coming from the load balancers and the allowed port range. The standard port range for Kubernetes load balancers (30000-32767) should be all that is needed here for ingress. If you are using transparent load balancers, you need to apply the ingress rules listed for load balancers above.

Egress

Port	Destination	Purpose
2376	VMs	Communication with VMs
4647	Nomad clients	Communication with the Nomad clients
all traffic	other nodes	Allow intra-cluster traffic

Nomad Clients

Nomad clients do not need to communicate with each other. You can block traffic between Nomad client instances completely.

Ingress

Port	Source	Purpose
4647	K8s nodes	Communication with Nomad server

Port	Source	Purpose
64535-65535	External	Rerun jobs with SSH functionality

Egress

Port	Destination	Purpose
2376	VMs	Communication with VMs
3000	VM Service load balancers	Internal communication
4647	Nomad Load Balancer	Internal communication
8585	Output Processor Load Balancer	Internal communication

External VMs

Similar to Nomad clients, there is no need for external VMs to communicate with each other.

Ingress

Port	Source	Purpose
22	Kubernetes nodes	Internal communication
22	Nomad clients	Internal communication
2376	Kubernetes nodes	Internal communication
2376	Nomad clients	Internal communication
54782	External	Rerun jobs with SSH functionality

Egress

You will only need the egress rules for internet access and SSH for your VCS.